

More Matplotlib

(CC-BY: Maria Skeppstedt)

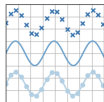
Matplotlib

A Python library for creating visualizations.

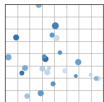
Example graphs:

Pairwise data

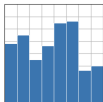
Plots of pairwise (x, y) , tabular $(var_0, \dots, var_n)$, and functional $f(x) = y$ data.



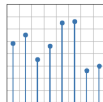
`plot(x, y)`



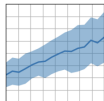
`scatter(x, y)`



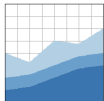
`bar(x, height)`



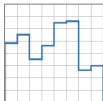
`stem(x, y)`



`fill_between(x, y1,
y2)`



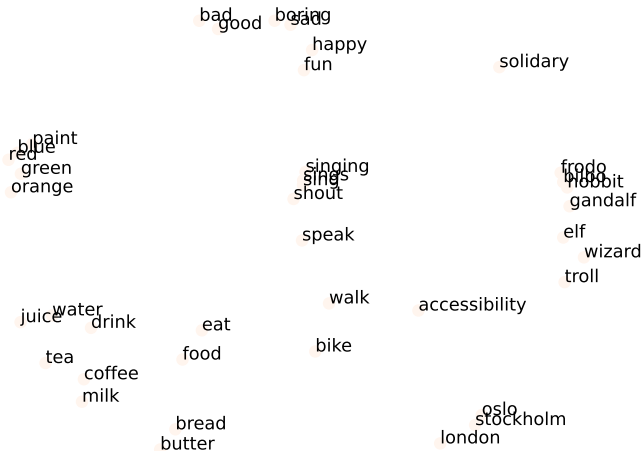
`stackplot(x, y)`



`stairs(values)`

Text annotations

TSNE(metric='cosine', random_state=1)



Text annotations

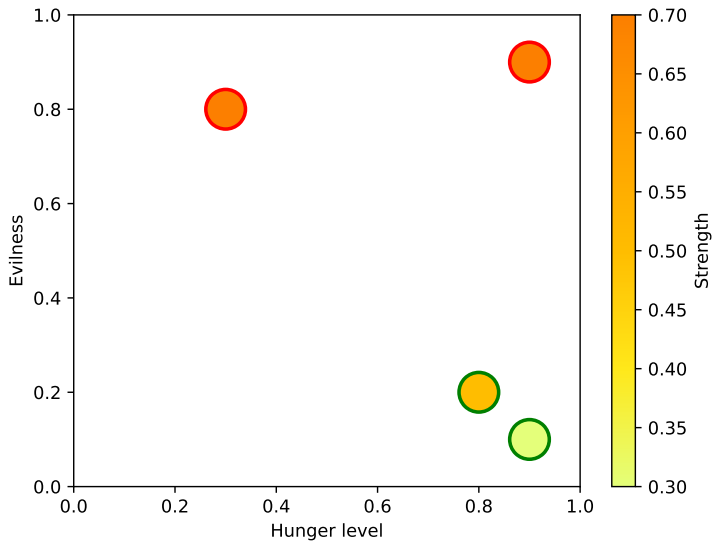
It is possible to annotate a scatter point with text

```
1 def plot_words(words, w_vect, dim_reductor, file_name):
2     plt.clf()
3     vector_space = \
4         np.array([w_vect.get_vector(w) for w in words])
5     two_dims = dim_reductor.fit_transform(vector_space)
6
7     for word, (x, y) in zip(words, two_dims):
8         plt.scatter(x, y, color="seashell")
9         plt.annotate(word, (x,y))
10
11     plt.axis('off')
12     plt.title(dim_reductor)
13     plt.savefig(file_name, format="pdf")
```

The plot can be saved to a file

Colormaps

Using colormaps



Colormaps

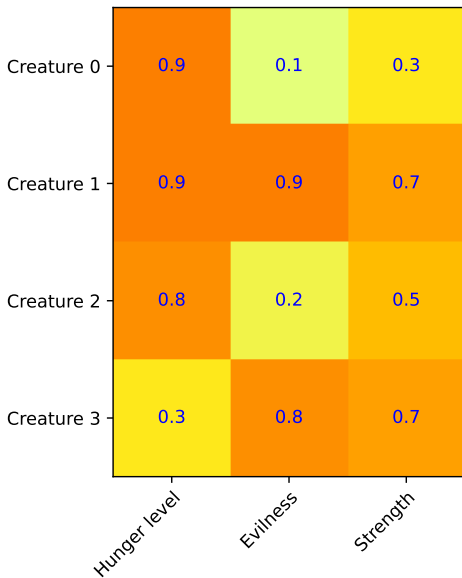
A color gradient is automatically created for the Strength values.

```
1 l_c = ["green", "red"]
2 features = ("Hunger level", "Evilness", "Strength")
3 X = np.array([[0.9, 0.1, 0.3],
4               [0.9, 0.9, 0.7],
5               [0.8, 0.2, 0.5],
6               [0.3, 0.8, 0.7]])
7 y = [0, 1, 0, 1]
8 # Plot hunger_level, Evilness
9 plt.scatter(X[:,0], X[:,1],\
10            edgecolor = [l_c[el] for el in y],
11            linewidth = 2,
12            s=500,
13            c=X[:,2],
14            cmap='Wistia')
15 plt.xlabel(features[0]); plt.ylabel(features[1])
16 plt.ylim(0,1); plt.xlim(0,1)
17 plt.colorbar(label='Strength')
18 plt.savefig('colormap.pdf')
```

The plot can be saved to a file

Heatmaps

Using colormaps



Heatmaps

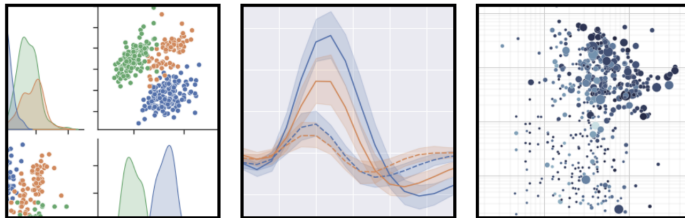
Code for the heatmap

```
1 features = ("Hunger level", "Evilness", "Strength")
2 X = np.array([[0.9, 0.1, 0.3],
3               [0.9, 0.9, 0.7],
4               [0.8, 0.2, 0.5],
5               [0.3, 0.8, 0.7]])
6 fig, ax = plt.subplots()
7 ax.imshow(X, cmap='Wistia')
8 ax.set_xticks(range(len(features)), labels=features,
9               rotation=45, ha="right", rotation_mode="anchor")
10 ax.set_yticks(range(X.shape[0]), \
11               [f"Creature {el}" for el in range(X.shape[0])])
12
13 print(X.shape)
14 for i in range(X.shape[0]):
15     for j in range(X.shape[1]):
16         text = plt.text(j, i, X[i, j],
17                         ha="center", va="center", color="b")
18         print(X[i, j])
19 fig.tight_layout()
20 plt.savefig('heatmap.pdf')
```

Seaborn

Seaborn is a Python library built on matplotlib

seaborn: statistical data visualization



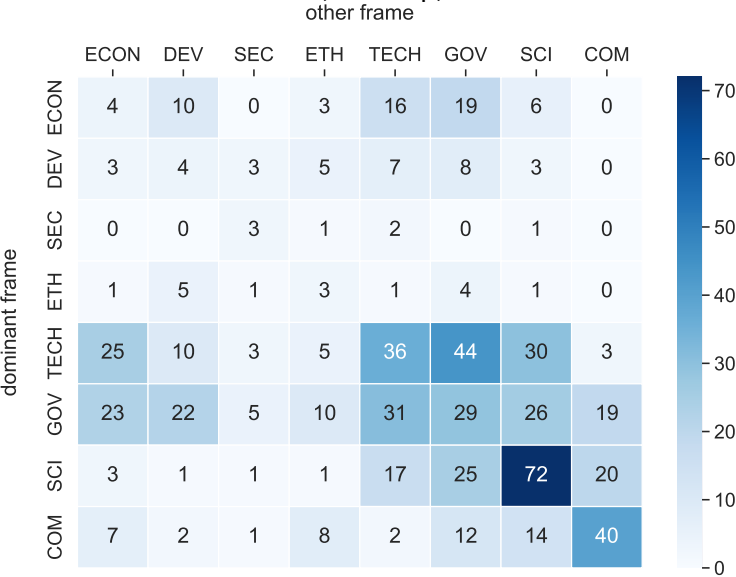
Seaborn is a Python data visualization library based on [matplotlib](#). It provides a high-level interface for drawing attractive and informative statistical graphics.

For a brief introduction to the ideas behind the library, you can read the [introductory notes](#) or the [paper](#). Visit the [installation page](#) to see how you can download the package and get started with it. You can browse the [example gallery](#) to see some of the things that you can do with seaborn, and then check out the [tutorials](#) or [API reference](#) to find out how.

To see the code or report a bug, please visit the [GitHub repository](#). General support questions are most at home on [stackoverflow](#), which has a dedicated channel for seaborn.

Seaborn

Seaborn confusion matrix (heatmap)



Seaborn confusion matrix (heatmap)

```
1 def plot_matrix(matrix, rows, columns, name,\n2     cmap, add_ylabel=True):\n3\n4     fig = plt.figure(figsize = (10,7))\n5     df_cm = pd.DataFrame(matrix, index = [i for i in rows],\n6                             columns = [i for i in columns])\n7     sn.set(font_scale=1.4)\n8\n9     h = sn.heatmap(df_cm, annot=True, \\\n10        cmap=cmap, linewidth = 1)\n11     h.set(xlabel = "other frame\\n")\n12     if add_ylabel:\n13         h.set(ylabel = "dominant frame\\n")\n14     else:\n15         h.set(ylabel = "    ")\n16     h.xaxis.set_ticks_position("top")\n17     h.xaxis.set_label_position("top")\n18     plt.savefig(name, format="pdf", \\\n19         bbox_inches='tight', pad_inches=0)\n20\n21 plot_matrix(label_matrix, classes, classes, \\\n22     "absolute.pdf", "Blues")
```

Seaborn confusion matrix (heatmap)

Reference:

Stede et al.: Framing climate change in Nature and Science editorials: applications of supervised and unsupervised text categorization in *Journal of Computational Social Science*